Local presents and the block: Brading's P1 and the coverage problem

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Abstract

Brading's 2015 formulation of presentism replaces P1 with P1, whose clause (a) says that each thing's present is less than the entire spatiotemporal evolution of the universe. She says that (a) is all that is needed to avoid the block universe. This paper argues that (a) is local boundedness and that local boundedness does not exclude the family of system-presents covering the whole history. The result is a source-grounded negative result: Brading's own stated sufficient condition does not secure a global existential restriction. It is not a refutation of presentism, a proof of eternalism, or a new theory of time. Nearby relativized views are distinguished, and the remaining repairs are stated with their costs.

1 Introduction

Brading's *Presentism as an Empirical Hypothesis* and its follow-up *Physically Locating the Present* offer one of the most careful attempts to keep a presentist ontology alive within special relativity (Brading, 2013, 2015). She rejects the shared assumption that spacetime is the ontological principle of unity of what exists, and replaces it with a law-constitutive principle: the dynamical laws constitute genuine unities, and the present of a system is the spatiotemporal region needed to sustain its dynamics.

Her 2013 paper states P1: for each thing, it exists only presently, where the spatiotemporal extent of that present is dynamics-dependent and empirically to be determined (Brading, 2013, p. 1110). She then notices that this extent might be the whole universe, in which case labeling the entire block "present" does not save what the presentist hoped to save. The 2015 paper therefore states P1:

> (P1) For each and every thing that exists, that thing exists only presently, where for each thing the spatiotemporal extent of the present for that thing is dependent on the dynamics and is (a) less than the entire spatiotemporal evolution of the universe as represented in Minkowski spacetime, and (b) less than the entire spatiotemporal evolution of that thing as represented in Minkowski spacetime (Brading, 2015, p. 17).

She adds: "In order to avoid the block universe, all we need is (a)" (Brading, 2015, p. 17).

This paper asks whether (a) does what she says it does. The answer defended here is negative and limited. Clause (a) is local boundedness: for every system i, its present $\mathcal{U}_i$ is a proper part of the world history $\mathcal{W}$. Local boundedness does not imply that the union of the $\mathcal{U}_i$ is a proper part of $\mathcal{W}$. If the actual system-presents cover $\mathcal{W}$, the account divides the history into dynamical

unities without excluding any event from existence. Brading's own sufficient condition therefore does not secure the global existential restriction.

This is a negative result about one published argument. It does not refute Brading's broader project, does not prove eternalism, and does not claim that every presentism fails. It also does not claim that the broader idea of system- or frame-relative existence is new: Balaguer's R-presentism (Balaguer, 2026) and Mason's domain- and level-dependent empirical presentism (Mason, 2026) already develop versions. The target here is narrower: Brading's P1 (a) and the claim that it suffices to avoid the block.

2 The existential contrast

2.1 What is being claimed

Brading begins from

- (P1) All and only things that exist now are real;
- (P2) Special relativity is a complete account of spatiotemporal structure (Brading, 2013, p. 1101).

P1 is a global restriction on existence: what there is is what is present, and events outside the present are not real. P2 says that nothing is missing from the special-relativistic account of spatiotemporal structure. The apparent conflict is that relativity supplies no global present.

Brading's response is not to add a preferred foliation. It is to reject the assumption that spacetime is the ontological ground of unity, and to ground unity in the dynamical laws instead (Brading, 2013, pp. 1104–1108). Spacetime becomes an epistemic principle of unity, the framework in which we organize knowledge beyond the here and now. The present becomes the region needed to sustain a dynamical system, and P1 is rewritten first as P1 and then as P1.

2.2 Existence, presence, unity and endurance

Four notions must be kept apart:

1. *Existence*: the claim that a thing or event is part of what there is.
2. *Presence*: a thing's or event's belonging to a system's sustaining region.
3. *Unity*: the law-constituted integration that makes a system one thing.
4. *Endurance*: the persistence of that unity through change.

Brading's argument moves from unity to presence. The further move from presence to a restriction on existence is the one at issue. P1 (b) is a claim about the extent of each thing's present—an endurance/presence claim—while (a) is a claim about the relation between each present and the whole history. Neither is yet a claim about the domain of existence.

2.3 The domain

For regimentation, let $\mathcal{W}$ be the block as represented in Minkowski spacetime, and for each genuine dynamical system i let $\mathcal{U}_i \subseteq \mathcal{W}$ be its system-sustaining present. This follows Brading's own phrasing in P1 (a): the comparison is made "as represented in Minkowski spacetime." The regimentation is a representational choice, not an ontological commitment. If Brading's fundamental entities are unities rather than events, the argument is to be read as a claim about the spatiotemporal extents of unities; the coverage point concerns the extents, not the category of the bearers.

Brading's 2013 footnote 7 denies determinate ontological spatiotemporal relations between systems (Brading, 2013, p. 1109). That denial is compatible with the representation: the block representation is the common frame in which the comparison is stated, not a claim that the systems stand in determinate ontological relations. If no common representation is allowed, the comparison becomes a comparison of frameworks; that is the deflationary fallback discussed in the conclusion.

3 Brading's law-constitutive present

Brading's positive proposal has three parts.

First, the laws constitute genuine unities. A physical thing, simple or composite, is what it is in part because it satisfies the laws; the laws also constitute the parts of a composite and their persistence through change (Brading, 2013, pp. 1105–1108). This is offered as an alternative to essentialism and haecceitism, and as an account of numerical identity through qualitative change.

Second, spacetime is demoted from an ontological to an epistemic principle of unity. It is the framework in which we organize knowledge of the not-here and not-now, not the ground of what exists (Brading, 2013, pp. 1108–1109).

Third, the present is the region needed to sustain the dynamical system:

> The present is a spatiotemporal region of whatever size is necessary to sustain the dynamical system in question (Brading, 2013, p. 1109).

The region may be much smaller than the block, or it may be the size of the block. If the only genuine system is the entire universe, Brading says that "this version of presentism is defeated and the block universe triumphs" (Brading, 2013, p. 1109). The empirical question is whether there are genuine subsystems whose sustaining regions are proper parts of the whole.

P1 states the first reformulation. Its full text is:

> (P1) For each and every thing, that thing exists only presently, where the spatiotemporal extent of that "present" is dependent on the dynamics, and it is something to be determined empirically (Brading, 2013, p. 1110); (Brading, 2015, p. 17).

Brading then observes that the extent might be the entire universe, so that "having the entirety of the block universe labeled as the 'present' doesn't seem to save anything that the presentist was hoping to save" (Brading, 2015, p. 17). P1 is the response. Clause (a) bounds each present relative to the universe; clause (b) bounds it relative to the thing's own history. Brading says (b) is optional for avoiding the block, and that "all we need is (a)."

A distinguished predecessor is Stein's event-relative notion of definiteness (Stein, 1991). Stein's printed theorem (p. 149) lists reflexivity, transitivity, invariance under time-orientation-preserving

automorphisms, and one past-pointing related pair, with the dimension qualification "of any number of dimensions—of course at least two." It is not self-contained: the proof also uses the non-universality condition (iv) and the prose premise that every event in the causal past—timelike or null—is definite as of the point (Stein, 1991, p. 149). The reflexive strict chronological past is non-universal but omits past-null pairs, so condition (iv) alone is not enough. The local-present strategy inherits this distinction; it does not inherit a global restriction.

4 The coverage objection

4.1 Local boundedness is not global restriction

Clause (a) says, for every system i,

$$\mathcal{U}_i \subsetneq \mathcal{W}.$$

It does not imply

$$\bigcup_i \mathcal{U}_i \subsetneq \mathcal{W}.$$

If every event belongs to at least one system-present, the family $\{\mathcal{U}_i\}$ covers $\mathcal{W}$. The account then divides the history into dynamical unities; it does not exclude any event from existence. The present can be "local, not global" (Brading, 2013, p. 1110) while the union of local presents is the whole history.

Proposition 1 (Non-implication)**.** *Local boundedness of every $\mathcal{U}_i$ is consistent with $\bigcup_i \mathcal{U}_i = \mathcal{W}$.*

Proof. Let $\mathcal{W}$ be a nonempty set and take two overlapping proper subsets whose union is $\mathcal{W}$. Every member is locally bounded, and the union is $\mathcal{W}$. □

The proposition establishes a non-implication, not that Brading's actual family covers $\mathcal{W}$. The further premise is:

(C) The actual system-presents cover $\mathcal{W}$.

If (C) holds, no event is excluded. If (C) fails, some event lies outside every system-present; that is a substantive no-coverage claim, not a consequence of P1 (a). Brading does not state (C) or its negation. The coverage objection is therefore exact as a non-implication and conditional as an objection: it shows that (a) does not by itself secure the restriction.

There are two textual routes to (C), both suggested by Brading's own text.

First, the law-constitutive approach applies to "the totality of existing things" as well as to simple and composite things (Brading, 2013, p. 1105, n. 3). If the totality is a genuine unity, its sustaining region is plausibly the whole history. Then its present is not less than the universe's evolution, contrary to (a), or the totality is not a thing to which (a) applies. Either way, P1 faces an internal pressure.

Second, Brading notes that she "can be made of things whose dynamically sustained regions of spacetime are much smaller than mine" (Brading, 2013, p. 1109, n. 6). A plurality of nested, short-lived things whose presents populate the history is exactly the kind of family for which (C) is natural. If their presents cover $\mathcal{W}$, local boundedness is satisfied and no event is excluded. This

is an argument sketch, not a proof; it is enough to show that (C) is not excluded by Brading's premises.

If (C) is not established, the conclusion is a burden-shift: (a) is insufficient unless a no-coverage premise is supplied. The paper should not call the objection "exact" without that qualification. What is exact is the non-implication.

4.2 The interaction passage

Brading's interaction passage is often read as a system-relative existence claim. In full:

> The ground of what is real is the dynamics, and we belong to the same world as whatever we interact with, and the rest of that world is as real as we are. ... Does my son exist relative to me when he is in London and I am here? Of course. There are plenty of interactions going on that link us. Is there a determinate fact of the matter about what he is doing right now? No. (Brading, 2013, p. 1110)

This passage is better used for the negative verdict than for a relativized existence attribution. "We belong to the same world," "the rest of that world is as real as we are," and "the unity of what there is, both of the parts and of the whole" are single-domain claims. "Exists relative to me" is glossed by interaction and followed by a claim about the *determinacy* of what the son is doing now, not about his existence. The passage supports the claim that no event is excluded merely by lying outside a local present; it does not establish a relativized existential quantifier. The paper therefore withdraws the stronger attribution.

Relativized existence is a real position, but it is not Brading's explicit P1. Balaguer's R-presentism makes existence relative to a frame of reference, with no frame-independent fact about which 3-manifold is real (Balaguer, 2026). Mason's empirical presentism makes the present domain- and level-dependent and treats "what exists" as perspective-relative (Mason, 2026). Those are nearby views; the present paper's target is P1 (a), and its conclusion is compatible with theirs.

5 Repairs and costs

If one wants the original global restriction, two repairs attempt to restore it.

Restoring repair 1: privileged actual systems. Only some systems' presents are actual. This restores a restriction, but the selection principle is not derived from the physics or from dynamical unity alone. Brading's own text supplies a candidate: the dynamics grounds "the unity of what there is, both of the parts and of the whole (consisting of interacting parts)" (Brading, 2013, p. 1110), and the law-constitutive approach applies to the totality. A maximal-interacting-unity principle is a dynamical selection principle. If the maximal unity is the selected system, its present is the whole history and the view yields the block. If it is not, the principle needs independent defense. Either way, the cost is real.

Restoring repair 2: restrictions on coexistence. Not all local presents can coexist. This may take the form of a maximal family of presents related by a stated coexistence relation, or a dynamics that selects one present. Without a formal relation, "maximal consistent set" is a placeholder; with one, the union of the family need not be a single present, and the view risks reintroducing a global present or a privileged foliation. The physics does not supply the relation.

Revisionary option 1: system-relative existence. Replace a single unrestricted existence predicate with "exists-for-system-i." This blocks the coverage objection by changing the question, but it gives up the global presentist claim. It is the route developed by Balaguer and Mason, not a restoration of P1.

Revisionary option 2: revised presentism. Give up global exclusion and accept local or functional becoming. This is coherent, but it is a different thesis: it may be an eternalist history with a local becoming, not presentism as the restriction on existence. If nothing distinguishes it from eternalism except the label, this is the paper's real conclusion.

The primitive-actuality reply deserves a hearing. A presentist may say that actuality is primitive and cannot be derived from the physics; the privileged system is simply the actual one. That is a coherent metaphysical posit. Its cost is that the physics does not supply it, and the paper's claim is only that P1 (a) does not supply it either. The reply does not restore the derivation; it relocates the premise.

6 A parity verdict

The conclusion is not that the evidence is silent. It is a parity verdict:

> On the P1 (a) reading, the physics does not favor presentism over eternalism; any advantage presentism can claim is purchased by an actuality or privilege premise the physics does not supply.

This is weaker than exclusion and stronger than a catalogue. A composite ontology can be favored without excluding every rival, but here the comparison yields a tie unless the actuality premise is added. Adding it is a metaphysical choice, not a physical result.

7 Strongest objections

Saunders: the equivalence-relation requirement. Saunders argues that presentism needs an invariant, definable equivalence relation of co-presence: "nothing less than an equivalence relation will do for the presentist"; "not only transitivity, but also symmetry, is required" (Saunders, 2000, p. 3). The definability results leave past-timelike, past-causal or universal relations; none is a non-trivial symmetric equivalence relation of a present (Clifton and Hogarth, 1995). The "non-trivial" qualifier is this paper's strengthening: the universal relation is an equivalence relation, so Saunders's bare "no equivalence relation" needs it.

A system-relative present is not an equivalence relation among all events; it is a region assigned to a system. Brading's view therefore either gives up the equivalence-relation requirement and with it the standard presentist claim, or needs an additional structure that defines co-presence. The coverage objection is the set-theoretic shadow of the first horn. Saunders's reason for the requirement—the present must be a public, non-provincial "whole of reality"—is not answered by a local present; if Brading rejects that starting point, she owes an account of why "present" is still a single public notion.

Pooley: taking both passage and relativity seriously. Pooley distinguishes two easy routes: a deflationary analysis of passage compatible with the block, which does not take passage seriously, and a privileged global present, which does not take relativity seriously (Pooley, 2013). His

hard route eschews a global Now and links passage to the open future. Brading's paper is about existence, not passage; she is not a candidate for Pooley's route. The honest formulation is that the coverage point is a *precondition* for any non-deflationary hard route: a non-deflationary structure is needed, and Brading's local present supplies a deflationary one unless an actuality premise is added. This is a concession, not a refutation of Pooley.

8 Conclusion

Brading's P1 is her strongest anti-block formulation. Clause (a) is local boundedness, and local boundedness does not imply a global existential restriction. If the actual system-presents cover the history, no event is excluded. Brading's "all we need is (a)" is therefore insufficient as stated. The conclusion is a limited, source-grounded negative result about one published argument.

The broader lessons are these. A local, law-constitutive present can be a serious account of unity and presence; it does not by itself yield a global ontology of existence. A presentist who wants the global restriction must add an actuality or privilege premise, restrict coexistence, or revise what presentism claims. The first is a metaphysical posit, the second risks a global present, and the third may be eternalism with a local becoming. The paper does not choose among them; it states the cost.

What would change the verdict. A proof that the actual system-presents cannot cover $\mathcal{W}$ would restore the restriction without an actuality premise. A physically motivated selection of privileged actual systems would supply the missing premise, though its cost would still need defense. A revised P1 that explicitly restricts the domain of existence, rather than the extent of each thing's present, would change the target. A demonstration that system-relative existence is the same global claim under another notation would defeat the revisionary route.

Limits. This is a philosophical argument, not an empirical prediction or an ontology selection. The coverage non-implication is elementary set theory; the contribution is the source-grounded application to P1 and the cost analysis. Brading 2013 and 2015 are read at full text (2015 pp. 17–18 in particular); Saunders and Pooley are read at the preprint depths stated in the review record; Balaguer and Mason are used at abstract/chapter level. No global priority or novelty claim is made beyond the modest P1 point identified in the accompanying novelty audit.

References

Mark Balaguer. *How to Be a Presentist*. Oxford University Press, 2026. doi: 10.1093/9780197845745.001.0001. Chapter: How to Make Presentism Consistent with Special Relativity.

Katherine Brading. Presentism as an empirical hypothesis. *Philosophy of Science*, 80(5):1101–1111, 2013. doi: 10.1086/673897.

Katherine Brading. Physically locating the present: A case of reading physics as a contribution to philosophy. *Studies in History and Philosophy of Science Part A*, 50:13–19, 2015. doi: 10.1016/j.shpsa.2014.09.007.

Rob Clifton and Mark Hogarth. The definability of objective becoming in Minkowski spacetime. *Synthese*, 103(3):355–387, 1995. doi: 10.1007/BF01089733.

Lucy Mason. Temporal perspectives, probabilities, and openness. *Synthese*, 2026. doi: 10.1007/s11229-026-05493-1. Preprint: PhilSci-Archive 25374 (2025).

Oliver Pooley. Relativity, the open future, and the passage of time. *Proceedings of the Aristotelian Society*, 113(3):321–363, 2013. doi: 10.1111/j.1467-9264.2013.00357.x. Preprint: PhilSci-Archive 9936.

Simon Saunders. Tense and indeterminateness. *Philosophy of Science*, 67(S3):S600–S611, 2000. doi: 10.1086/392849. Preprint: PhilSci-Archive 2241.

Howard Stein. On relativity theory and openness of the future. *Philosophy of Science*, 58(2): 147–167, 1991. doi: 10.1086/289609.